NUFYP Mathematics
Spring Semester

Learning Block 2

2.3 Probability 2

NAZARBAYEV UNIVERSITY

Learning outcomes

2.3.1 Describe the difference between independent and dependent events.

2.3.2 Apply the Multiplication Rule to solve “and” probability problems involving *independent events*

2.3.3 Apply the Multiplication Rule to solve “and” probability problems involving *dependent events* (conditional probability)

2.3.4 Use a Venn diagram to solve conditional probability problems

2.3.5 Use a tree diagram to solve probability problems

Lecture Outline

Independent events

- Multiplication Rule

Dependent Events (conditional probability)

- Using the Multiplication Rule
- Using a tree diagram
- Using a Venn diagram

Introduction

In the last lecture, we found simple probabilities as well as the probability that **event A OR event B** occurs, written $P(A \cup B)$.

In this lecture we will learn how to find the probability that **event A AND event B** occur, written $P(A \cap B)$. We saw basic examples of $P(A \cap B)$ in the last lecture, but here we will look closer at the probability of two events both occurring.

To this end, we need to consider whether events A and B have any relationship to each other. In other words, does the occurrence of one affect the probability that the other will happen? If there is **no dependence relationship** between events A and B, they are said to be **independent events**. If there is a **dependence relationship** between events A and B, they are said to be **dependent events**.

Independent vs Dependent events

Examples of **independent events**:

1) Flipping a coin two times and asking what is the probability of getting *heads* on the first flip **and** *tails* on the second flip. These two events are completely independent of each other. That is, the occurrence of one does not affect the probability of the other.

2) Rolling a die two times and asking what is the probability of getting a 5 on the first roll **and** a 6 on the second roll. Again, what comes up on one roll of a die in no way affects what comes up on another roll, so these two events are independent of each other.

Independent vs Dependent events

Example of **dependent events**:

What is the probability that it snows tomorrow **and** class is canceled tomorrow? Clearly there is a dependence relationship between these two events. That is, the probability that class is canceled tomorrow is different *depending on* whether it snows tomorrow.

Multiplication Rule ("and" probabilities)

Independent Events

For independent events, the probability of A **and** B is:

$$\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B}) = \mathbf{P}(\boldsymbol{A}) \times \mathbf{P}(\boldsymbol{B})$$

Here, the probability that A and B both happen is simply the product of their two individual probabilities. It makes sense that this would be the case since the two events do not affect each other.

Example 1

Events A and B are independent and

$P(A) = 1/3$ and $P(B) = 1/5$.

Find

a) $P(A \cap B)$

b) $P(A \cap B')$

Solution

a) $P(A \cap B) = P(A) \times P(B) = 1/3 \times 1/5 =$ **1/15**

b) If *A* and *B* are independent, then so are A and B'.

Therefore, $P(A \cap B') = P(A) \times P(B')$

Remember that $P(B') = 1 - P(B) = 1 - 1/5 = 4/5$

So, $P(A \cap B') = 1/3 \times 4/5 =$ **4/15**

Your turn!

Events A and B are independent and

$P(A) = 1/3$ and $P(B) = 1/5$.

Find $P(A' \cap B')$.

Solution

If *A* and *B* are independent, A' and B' must also be independent.

Therefore $P(A' \cap B') = P(A') \times P(B')$

and $P(A') = 1 - P(A) = 1 - 1/3 = 2/3$

so, $P(A' \cap B') = 2/3 \times 4/5 =$ **8/15**

Multiplication Rule ("and" probabilities)

Dependent Events

For dependent events, the probability of A **and** B is:

$$\mathbf{P(\boldsymbol{A} \cap \boldsymbol{B}) = P(\boldsymbol{B}) \times P(\boldsymbol{A} \mid \boldsymbol{B})}$$

In this formula we have a new probability, P(*A* | *B*), and this is where the *dependence relationship* comes into play. P(*A* | *B*) is the probability that A happens given that B has happened. It is read "the probability of A given B."

Such probabilities are called **conditional probabilities**.

Conditional Probability

We can solve for P(A |B) in the formula

$$\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B}) = \mathbf{P}(\boldsymbol{B}) \times \mathbf{P}(\boldsymbol{A} \mid \boldsymbol{B})$$

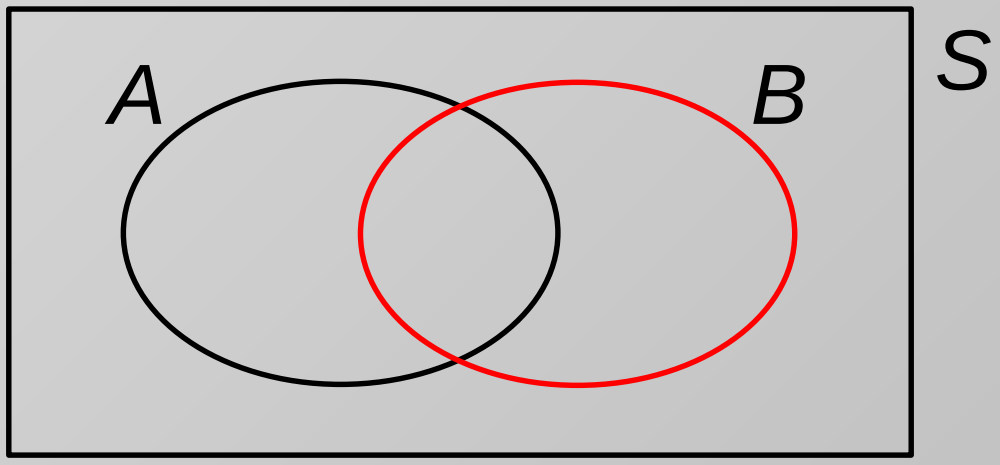
to derive the **conditional probability formula**:

$$\mathbf{P}(\boldsymbol{A}|\boldsymbol{B}) = \frac{\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B})}{\mathbf{P}(\boldsymbol{B})}$$

But let’s also see why this formula makes sense.

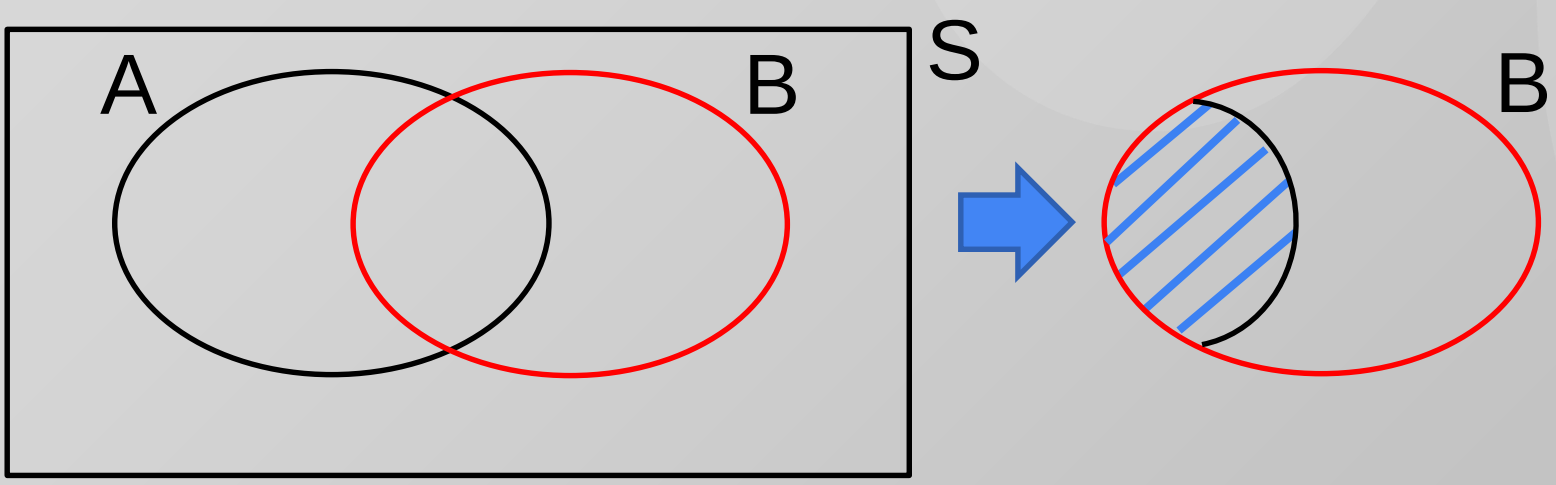
Conditional Probability

We can make sense of the **conditional probability formula** for $P(A|B)$ using a Venn diagram:



Since event B has already occurred, our sample space is reduced from S to B.

We are interested in the probability of A given B. So, locate the area for A on the Venn diagram within B.



This shaded area is A ∩ B.

Remember that probability is found by P(desired outcomes)/P(total)

For us, $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Conditional Probability

It is worth noting that since $\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B}) = \mathbf{P}(\boldsymbol{B} \cap \boldsymbol{A})$,

we have that

$$\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B}) = \mathbf{P}(\boldsymbol{B}) \times \mathbf{P}(\boldsymbol{A} \mid \boldsymbol{B}) = \mathbf{P}(\boldsymbol{A}) \times \mathbf{P}(\boldsymbol{B} \mid \boldsymbol{A})$$

and so

$$\mathbf{P}(\boldsymbol{A}|\boldsymbol{B}) = \frac{\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B})}{\mathbf{P}(\boldsymbol{B})} \quad \text{and} \quad \mathbf{P}(\boldsymbol{B}|\boldsymbol{A}) = \frac{\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B})}{\mathbf{P}(\boldsymbol{A})}$$

Example 2

C and D are two events such that $P(C) = 0.2$, $P(D) = 0.6$ and $P(C|D) = 0.3$.

Find

a) $\mathbf{P(D|C)}$

b) $\mathbf{P(C' \cap D')}$

c) $\mathbf{P(C' \cap D)}$

Solution

a) Using the conditional probability formula:

$$P(D|C) = P(C \cap D)/P(C)$$

We need to find $P(C \cap D)$.

Using the multiplication rule,

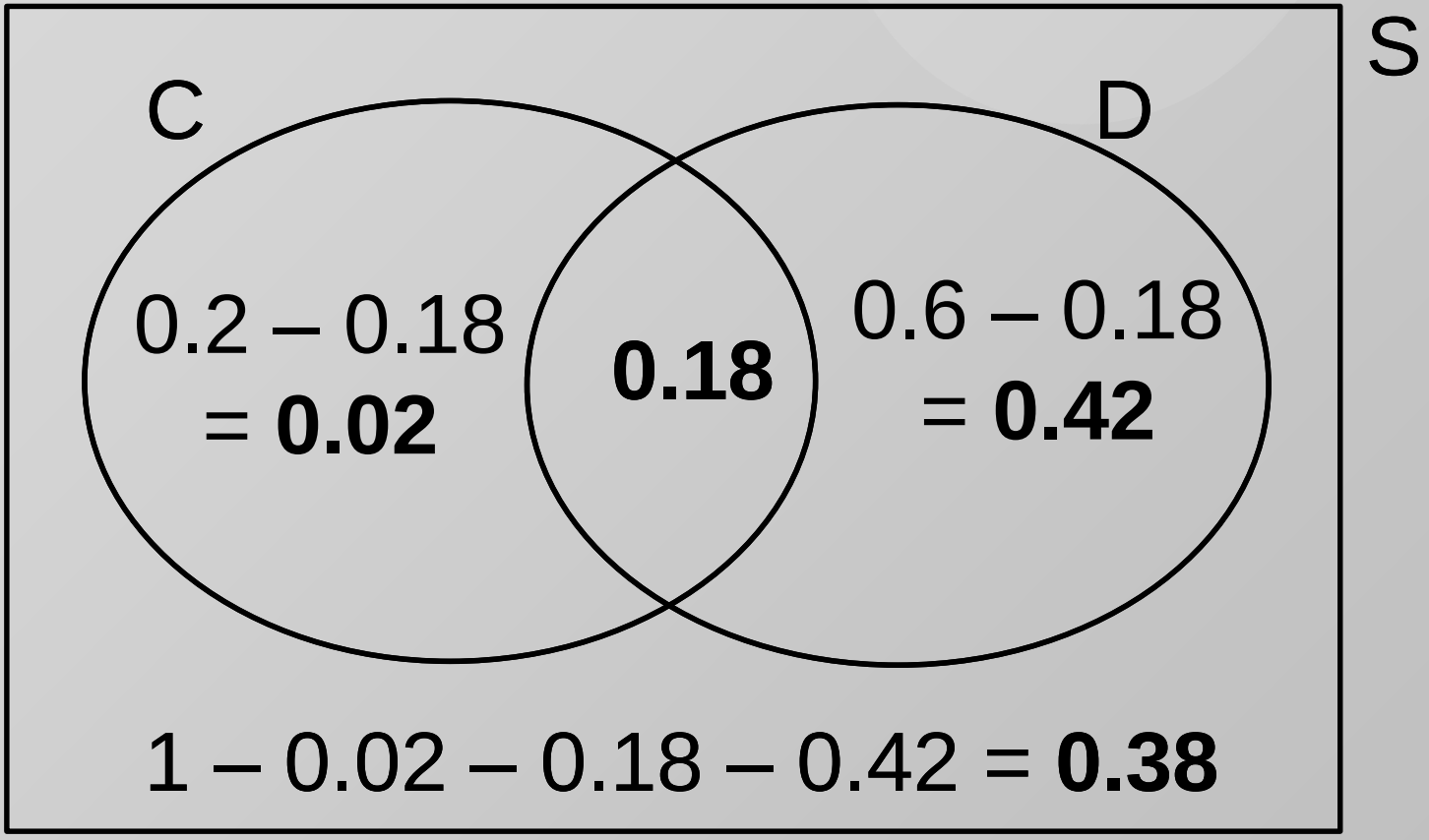
$$P(C \cap D) = P(C|D) \times P(D)$$
$$= 0.3 \times 0.6 = 0.18$$

Now $P(D|C) = P(C \cap D)/P(C)$

$$= 0.18/0.2 = \mathbf{0.9}$$

Solution

For parts b) and c) we will use a Venn diagram.

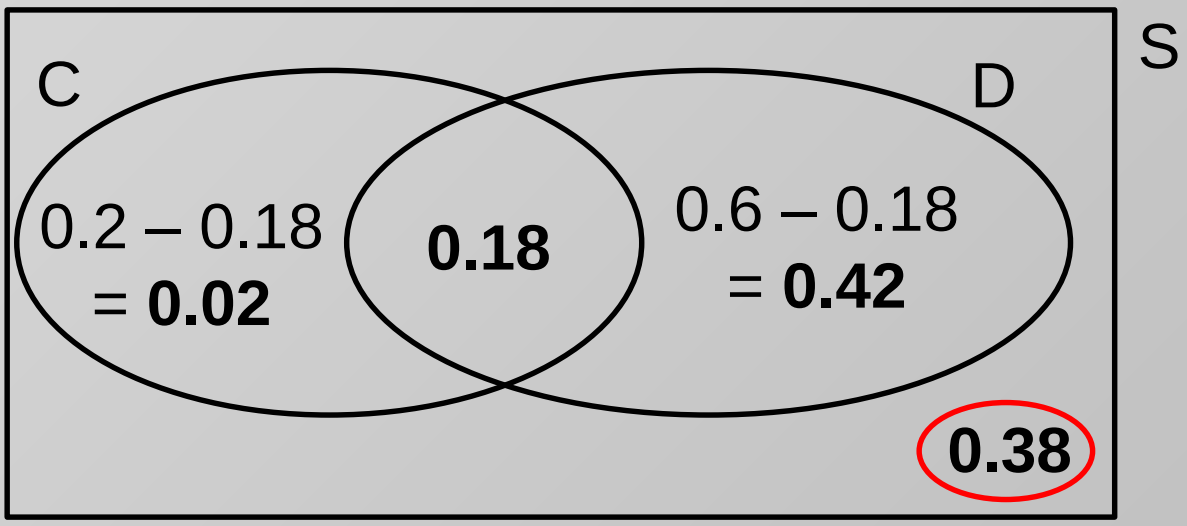


Solution

b) P(*C*'∩*D*')

This means *not* in C ***and*** *not* in D.

P(C' ∩ D') = **0.38**

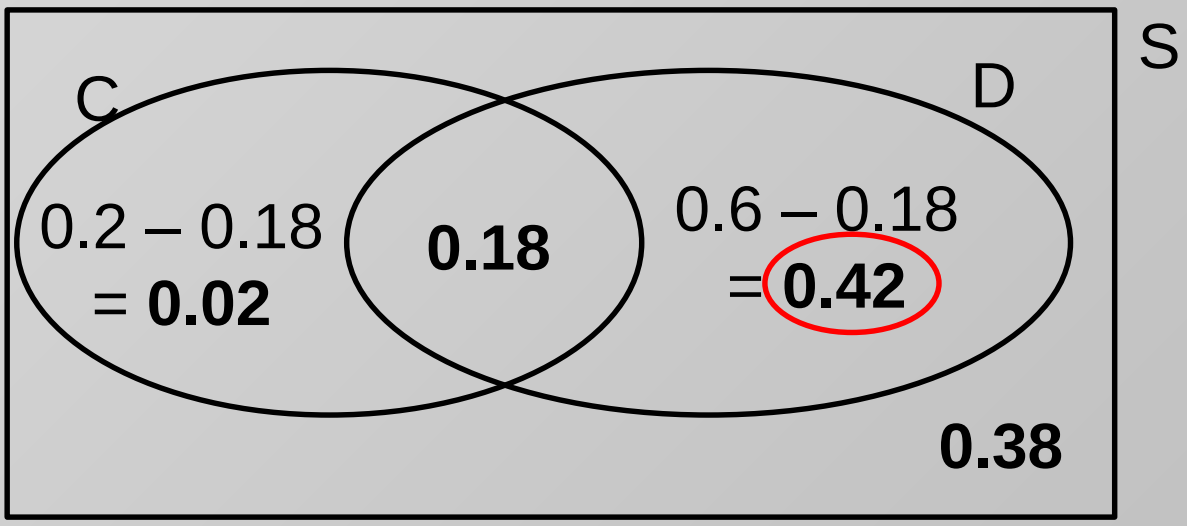


Solution

c) P($C'\cap D$)

This means *not* in C ***and*** in D.

P(C' ∩ D) = **0.42**



A tree diagram

A tree diagram is used to show all possible outcomes of a sequence of events.

Example 3

The probability that Adil spends more than four hours between 11.00 pm and 7.00 am playing computer games is 0.7. If he does do this the probability that he fails to attend his 9.00 am seminar is 0.9. If he doesn't do this the probability that he misses his 9.00 am seminar is 0.2. *Represent this information using a tree diagram.*

Solution

The probability that Adil misses his 9.00 am seminar or not is dependent upon whether he plays computer games for more than four hours during the night. This is a **conditional probability** situation.

Let A represent the event that Adil plays computer games for more than four hours.

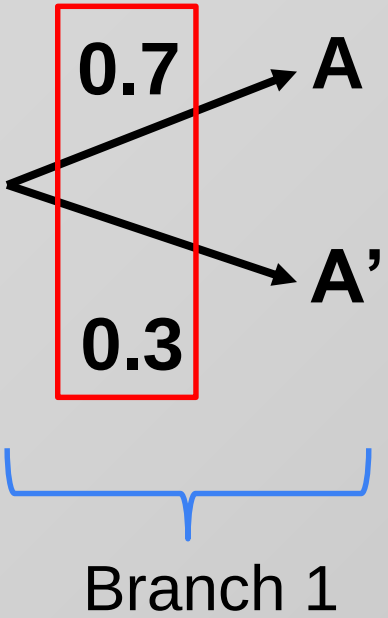
Let B represent the event that Adil misses his 9.00 am seminar.

Going back to the question,

- the probability of missing a seminar (Event B) ***if*** playing games more than 4 hours (Event A) is 0.9. That is $P(B|A) = 0.9$.

- the probability of missing a seminar (Event B) ***if not*** playing games more than 4 hours (Event A') is 0.2. That is $P(B|A') = 0.2$.

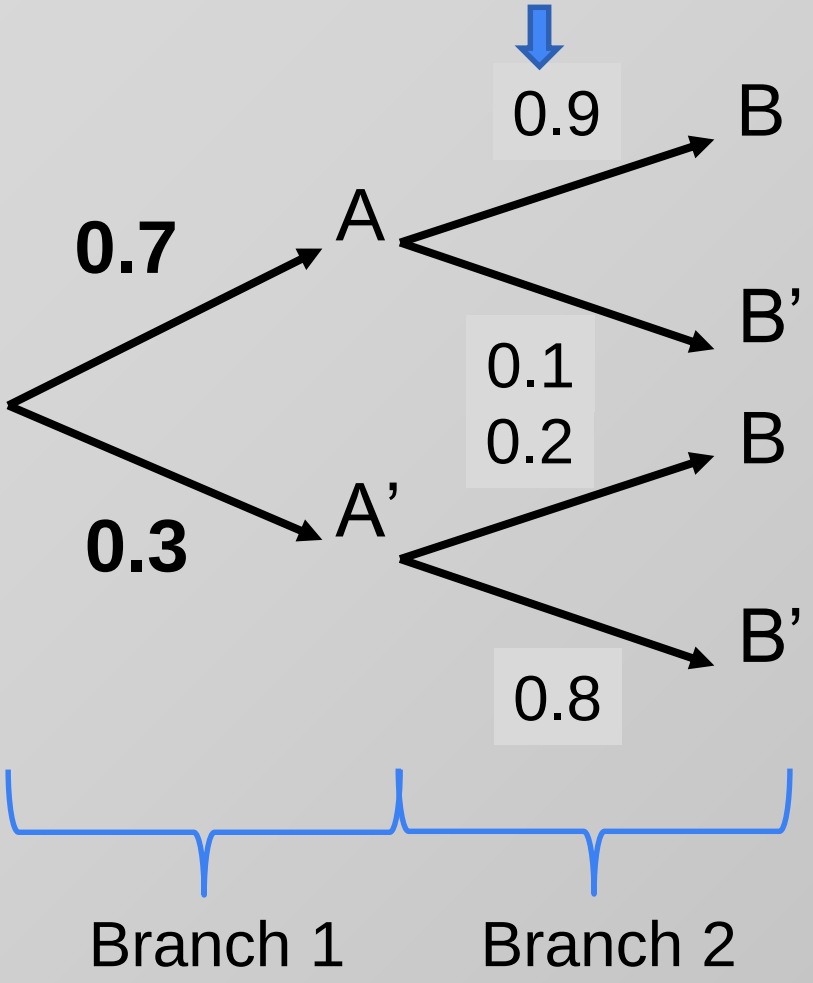
Let's consider the first event, A. Because Adil either plays computer games for more than four hours or he doesn't, we know that $P(A') = 1 - P(A) = 1 - 0.7 = 0.3$



We will make the second branch of the tree diagram on the next slide, which will refer to event B.

Sum of the probabilities from the same root always equals 1 = 100%.

These branches correspond to the conditional probabilities because either A happens or not is assumed.



? = P(B|A)

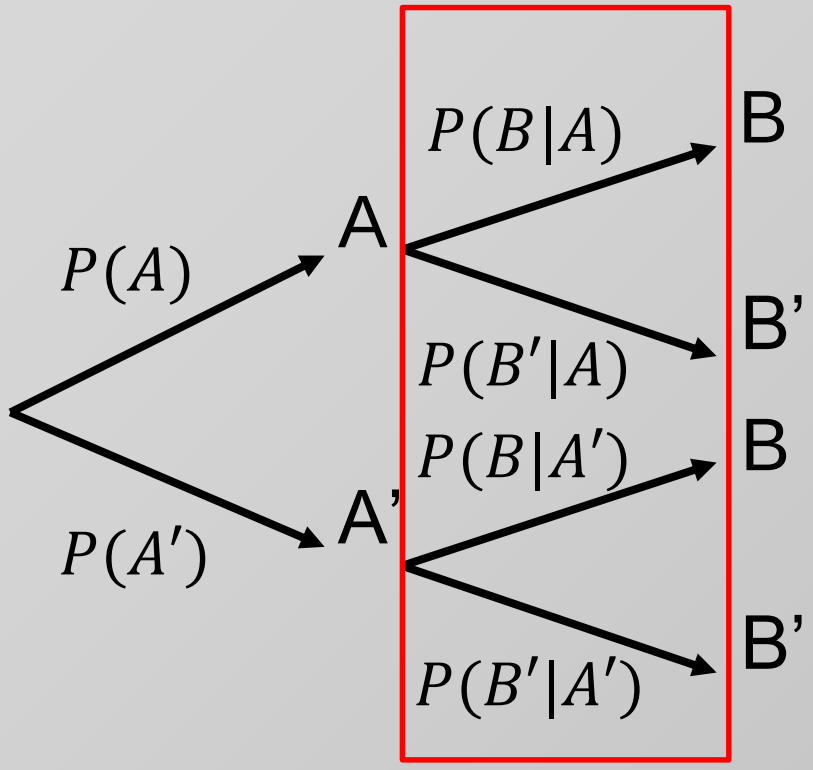
? = P(B'|A)

? = P(B|A')

? = P(B'|A')

Can you complete the tree diagram by identifying the four missing probabilities?

The conditional probability $P(B|A)$ and $P(A \cap B)$ are different!!!!!



The multiplication rule says,

$$P(A \cap B) = P(A) \times P(B|A)$$

Hence, in order to obtain $P(A \cap B)$, we multiply the numbers following the appropriate routes.

What about $P(A' \cap B)$?

These are conditional probabilities!

Example (continued)

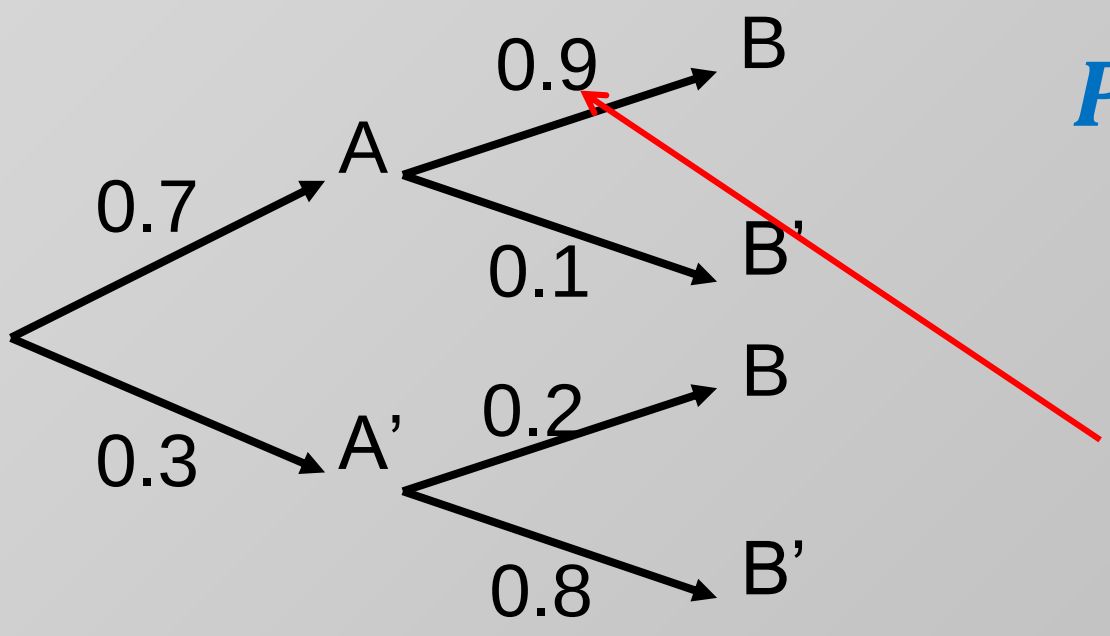
From the tree diagram in the previous slide, find $P(A \cap B)$, $P(A \cap B')$, $P(A' \cap B)$ and $P(A' \cap B')$.

What is the sum of all these four probabilities?

The sum is 1!

Example (continued)

What is the probability that Adil played a game for more than 4 hours (*event A*) on a certain day given that he missed the 9 am seminar next day (*event B*)?



$$\boldsymbol{P(A|B)} = ?$$

This is $P(B|A)$, not $P(A|B)$

Solution

By definition,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = 0.7 \times 0.9 = 0.63$$

$$P(B) = P(A \cap B) + P(A' \cap B)$$

$$= 0.7 \times 0.9 + 0.3 \times 0.2 = 0.69$$

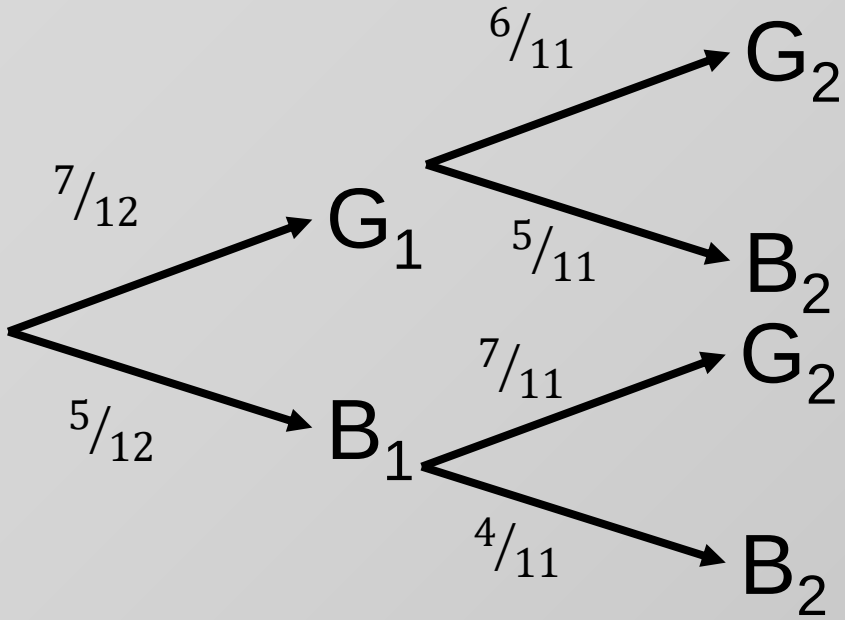
Therefore, $P(A|B) = \frac{0.63}{0.69} = \frac{63}{69}$

Example 4

A bag contains seven green beads and five blue beads. A bead is taken from the bag at random, the color is recorded and it is not replaced. A second bead is then taken from the bag and its color is recorded.

Find the probability that one bead is green and another bead is blue.

Solution



It can be (G_1 and B_2) OR (B_1 and G_2)

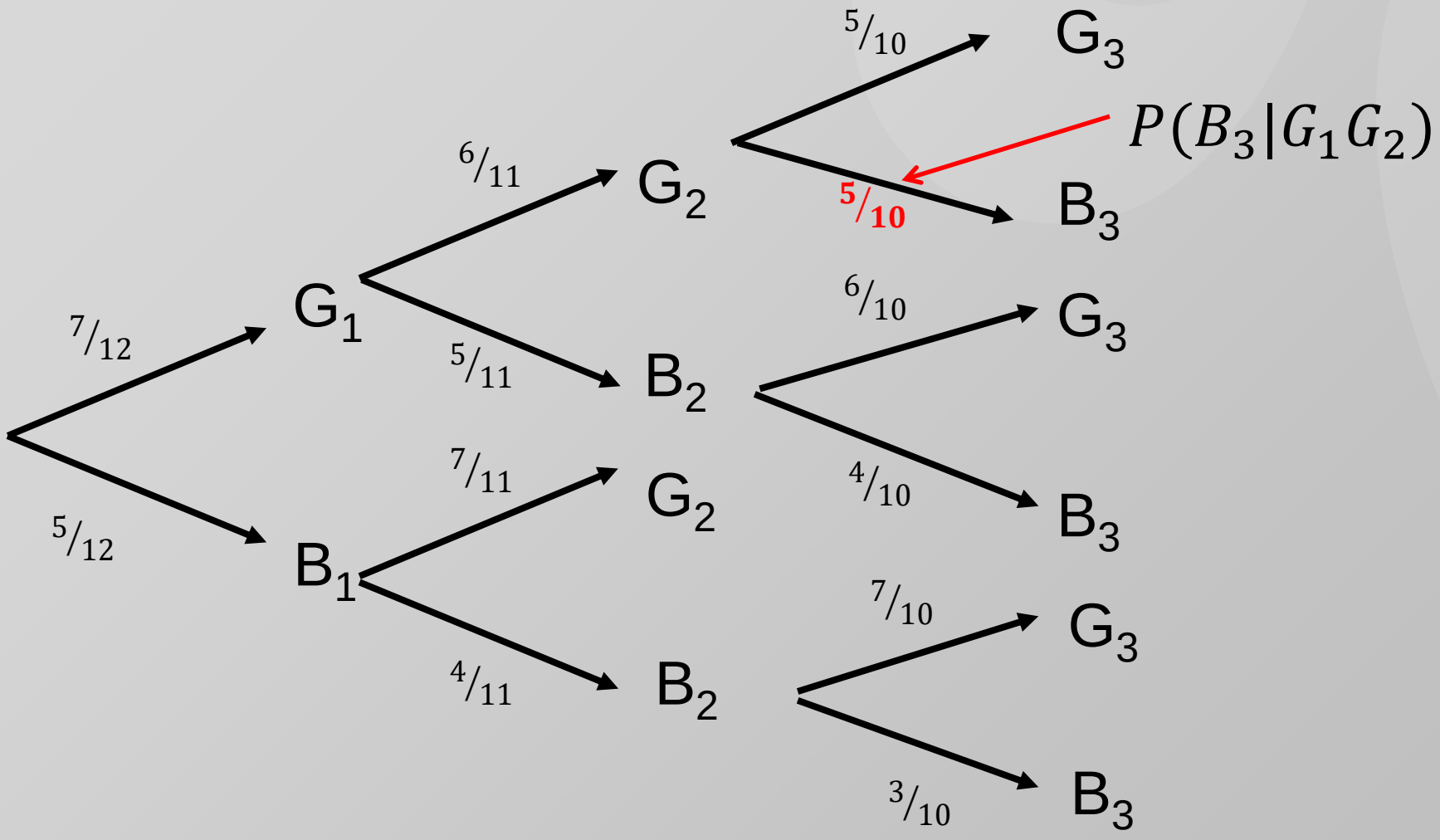
$$P(G_1 \cap B_2) + P(B_1 \cap G_2) = \frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11} = \frac{70}{132} = \frac{35}{66}$$

Your turn!

A bag contains seven green beads and five blue beads. A bead is taken from the bag at random, the color is recorded and it is not replaced. A second bead is then taken from the bag and its color is recorded and it is not replaced. A third bead is taken from the bag and its color is recorded.

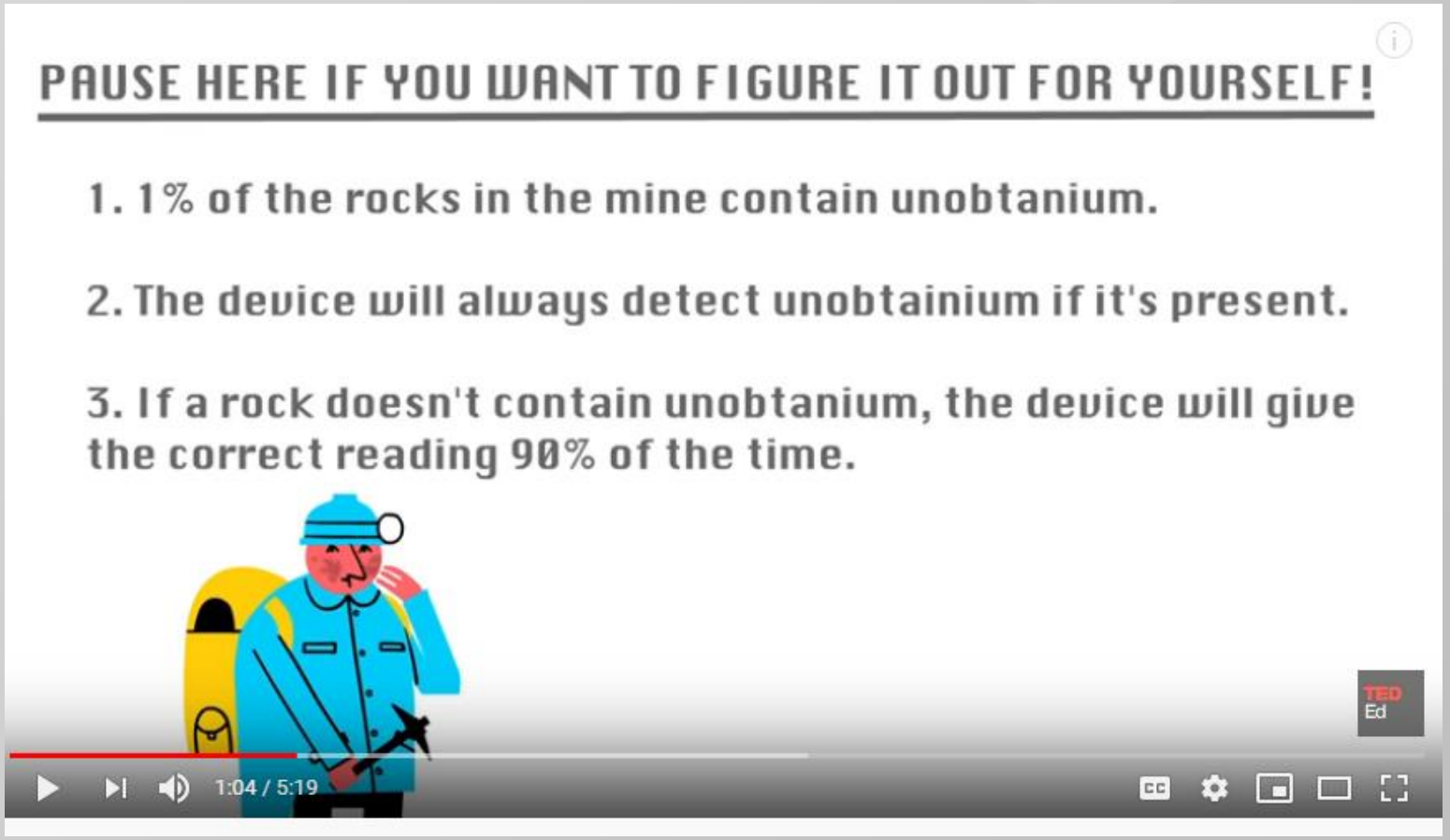
Find the probability that the third bead is blue given that the first two were green.

Solution


7/12
5/12
G1
B1
6/11
5/11
7/11
4/11
G2
B2
G2
B2
5/10
5/10
6/10
4/10
7/10
3/10
G3
B3
G3
B3
G3
B3
P(B3|G1G2)

Your turn!

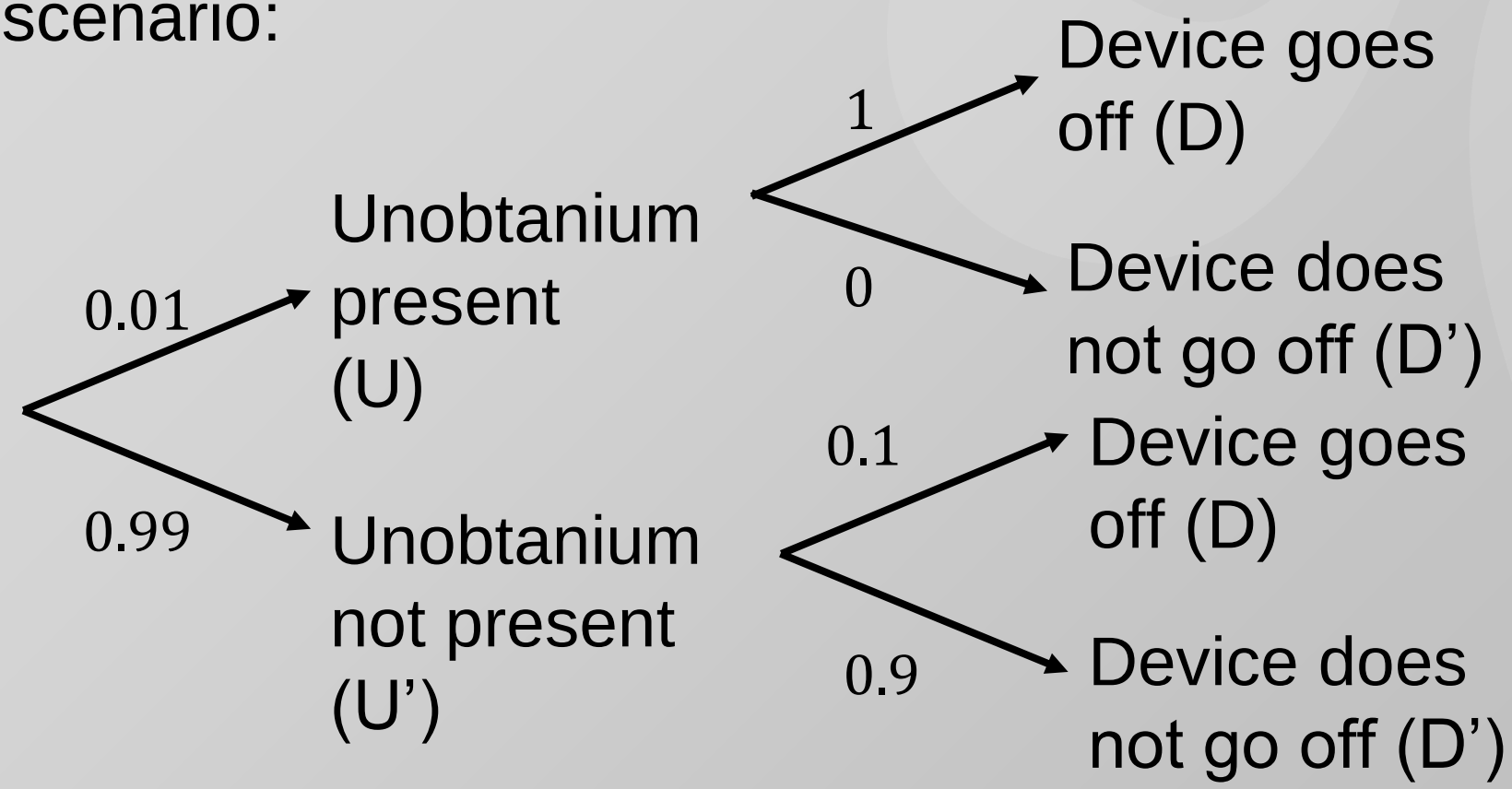
Watch this fun video until you see the screen below (1:04), then try to solve the riddle!!
https://www.youtube.com/watch?v=1csFTDXXULY



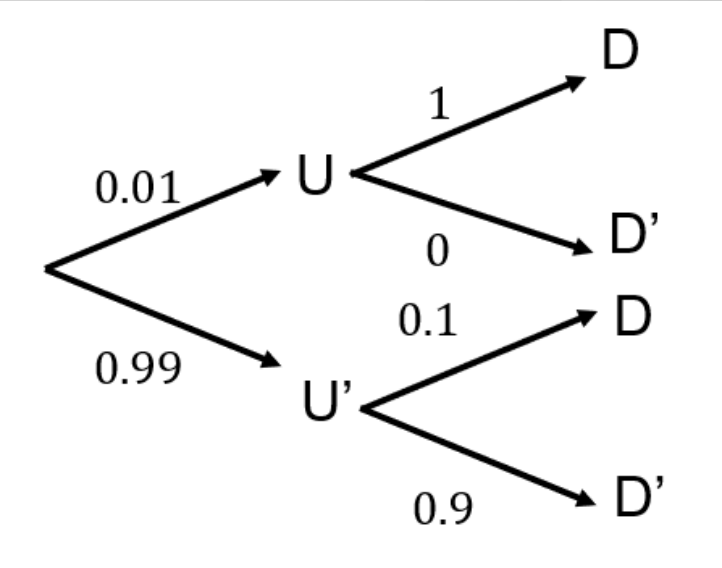
The device signals that a rock in the cart contains unobtanium. Your friend offers to sell you the rock for $200. If it is unobtanium, it will be worth $1000.

What is the probability that the rock contains unobtanium (given that the device goes off)? Should you make the trade?

Let's draw a tree diagram to represent the given scenario:



We want to find the probability that a random rock contains unobtanium given that the device goes off.



That is,

$$P(U|D) = \frac{P(U \cap D)}{P(D)}$$

$$= \frac{P(U \cap D)}{P(U \cap D) + P(U' \cap D)}$$

$$= \frac{0.01 \times 1}{0.01 \times 1 + 0.99 \times 0.1} = \frac{10}{109}$$

If the rock in the cart contains unobtanium, you will earn $1000.

The ‘expected’ value of the rock is the probability that the rock contains unobtanium multiplied by $1000.

$$P(U|D) \times \$1000 = \frac{10}{109} \times \$1000 \approx \$91.7$$

Your expected earning ($91.7) is a lot less than the cost ($200).

You should not make the deal!

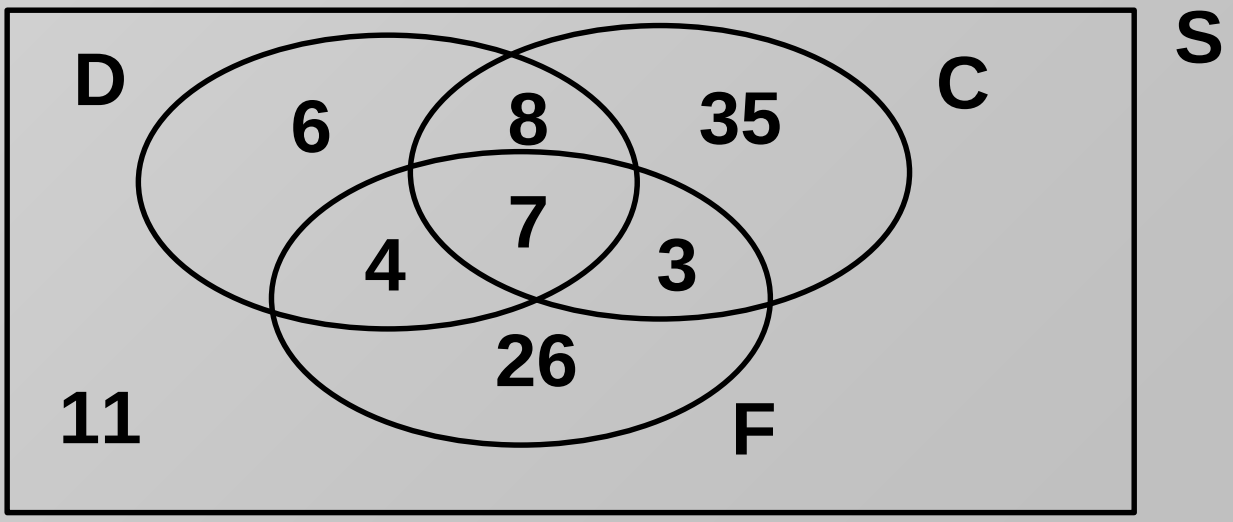
Conditional probability using a Venn diagram

Example 5 Do you remember the three event Venn diagram from the first probability lecture (example 8)? Use the Venn diagram to find the following conditional probabilities for a randomly chosen client.

a) own cats given that they own dogs

b) own cats given they own dogs and tropical fish

c) own cats given they own dogs or tropical fish or both.



Solution

a) Since owning a dog is already assumed, you need to restrict your interest to “D” only and the total number of outcomes is 6+4+7+8=25.
In the region for “D”, 8+7=15 also own a cat.

Therefore, $P(C|D) = \frac{15}{25} = 0.6$

b) Similarly, $P(C|D \cap F) = \frac{7}{11}$

c) $P(C|D \cup F) = \frac{8+7+3}{6+4+8+7+3+26} = \frac{18}{54} = \frac{1}{3}$

Connecting the Addition Rules

Recall how we can *always* use the following **Addition Rule** for *any* two events A and B:

$$\mathbf{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

However, when A and B are ***mutually exclusive***, this formula simplifies to

$$\mathbf{P(A \cup B) = P(A) + P(B)} \quad \text{since} \quad \mathbf{P(A \cap B) = 0}$$

Connecting the Multiplication Rules

The same is true for the Multiplication Rule! We can *always* use the following **Multiplication Rule** for *any* two events A and B: $\mathrm{P}(A \cap B) = \mathrm{P}(B) \times \mathrm{P}(A|B)$

However, if A and B are **independent**, then the probability of A is the same whether B has happened or not.

This means that **if A and B are independent** then, $\mathrm{P}(A|B) = \mathrm{P}(A)$ which gives

$$\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$$

Example 6

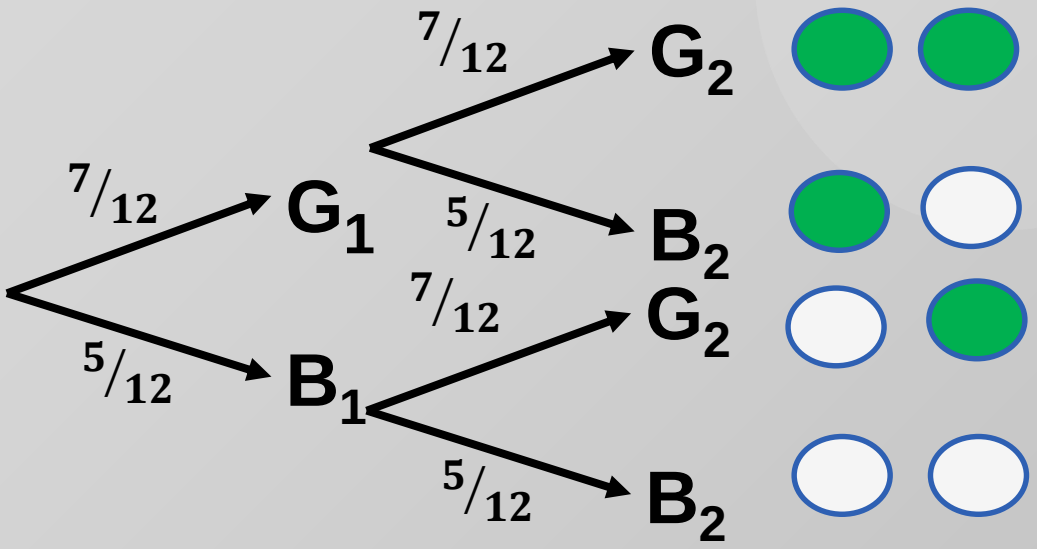
A bag contains seven green beads and five blue beads. A bead is taken from the bag at random, the color is recorded and the bead is replaced. A second bead is then taken from bag and its color recorded.

Draw a tree diagram and then answer the following:

a) Find the probability that one bead is green and another bead is blue.

b) Since we are replacing the bead from our first pick, it might be intuitively clear that the events *the first bead is green* and *the second bead is green* are independent. Show that the multiplication rule $P(A \cap B) = P(A)P(B)$ holds for these two events.

Solution



a) $P(G_1 \cap B_2) + P(B_1 \cap G_2) = \frac{7}{12} \times \frac{5}{12} + \frac{5}{12} \times \frac{7}{12} = \mathbf{\frac{35}{72}}$

Solution (continued)

b) We want to show that

$$P(G_1 \cap G_2) = P(G_1) \times P(G_2)$$

$P(G_1) = \frac{7}{12}$ from the tree diagram.

What is $P(G_2)$?

The first bead can be either green or blue.

That is, (G_1 and G_2) or (B_1 and G_2).

Solution (continued)

$$P(G_2) = P(G_1 \cap G_2) + P(B_1 \cap G_2)$$

$$= \frac{7}{12} \times \frac{7}{12} + \frac{5}{12} \times \frac{7}{12} = \mathbf{\frac{7}{12}}$$

$P(G_1 \cap G_2) = \frac{7}{12} \times \frac{7}{12}$ and $P(G_1) = P(G_2) = \frac{7}{12}$.
Hence, $P(G_1 \cap G_2) = P(G_1) \times P(G_2)$ which confirms that G_1 and G_2 are independent.

Formulae

- Multiplication rule for *independent events*

$$\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B}) = \mathbf{P}(\boldsymbol{A}) \times \mathbf{P}(\boldsymbol{B})$$

- Multiplication Rule for *dependent events*

$$\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B}) = \mathbf{P}(\boldsymbol{B}) \times \mathbf{P}(\boldsymbol{A}|\boldsymbol{B})$$
$$= \mathbf{P}(\mathbf{A}) \times \mathbf{P}(\boldsymbol{B}|\boldsymbol{A})$$

- Conditional probability

$$\mathbf{P}(\boldsymbol{A}|\boldsymbol{B}) = \frac{\mathbf{P}(\boldsymbol{A} \cap \boldsymbol{B})}{\mathbf{P}(\boldsymbol{B})}$$

Learning outcomes

2.3.1 Describe the difference between independent and dependent events.

2.3.2 Apply the Multiplication Rule to solve “and” probability problems involving *independent events*

2.3.3 Apply the Multiplication Rule to solve “and” probability problems involving *dependent events* (conditional probability)

2.3.4 Use a Venn diagram to solve conditional probability problems

2.3.5 Use a tree diagram to solve probability problems

Preview: Binomial expansions

Expansion of $(a+b)^n$

- n positive integer
  - Finite series
- n any other rational number
  - Infinite series
  - Needs a convergence condition